Mathematical Statistics I

Lecture notes based on Kim, *Mathematical Statistics*, Ch. 1–5

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Chapter 1. Probability Distributions

1.1 Probability and its properties

A **sample space** S is the set of all outcomes of an experiment; **events** are subsets of S. A probability P assigns numbers to events subject to Kolmogorov's axioms:

(A1) $P(A) \geq 0$ for every event A;

(A2) $P(S) = 1$;

(A3) (countable additivity) if $A_1, A_2, \ldots$ are pairwise disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem 1.1

(a) $P(\emptyset) = 0$, $0 \leq P(A) \leq 1$, and $P(A^c) = 1 - P(A)$.

(b) (Monotonicity) $A \subseteq B \Rightarrow P(A) \leq P(B)$.

(c) (Inclusion–exclusion) $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{n-1} P(A_1 \cap \cdots \cap A_n)$.

(d) (Countable subadditivity) $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (a) Apply (A3) to $S = S \cup \emptyset \cup \emptyset \cup \cdots$ to get $P(\emptyset) = 0$, then to the disjoint union $S = A \cup A^c$. (b) $B = A \cup (B \setminus A)$ disjointly. (c) For $n = 2$, $A_1 \cup A_2 = (A_1 \setminus A_2) \cup (A_2 \setminus A_1) \cup (A_1 \cap A_2)$ and $A_k = (A_k \setminus A_j) \cup (A_1 \cap A_2)$; combine. The general case follows by induction on n. (d) Let $B_1 = A_1$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. The B_n are disjoint with the same union and $B_n \subseteq A_n$, so $P(\bigcup A_i) = \sum P(B_n) \leq \sum P(A_n)$. ∎

Theorem 1.2 (Continuity of probability)

If $A_1 \subseteq A_2 \subseteq \cdots$ then $P(\bigcup_n A_n) = \lim_n P(A_n)$; if $B_1 \supseteq B_2 \supseteq \cdots$ then $P(\bigcap_n B_n) = \lim_n P(B_n)$.

Proof. With $C_1 = A_1$, $C_n = A_n \setminus A_{n-1}$ (disjoint, same union),

$$P\Big(\bigcup_n A_n\Big) = \sum_k P(C_k) = \lim_n \sum_{k \leq n} P(C_k) = \lim_n P(A_n).$$

. For decreasing B_n, apply this to $A_n = B_n^c$ and take complements. ∎

For example, under the uniform probability $P((a, b]) = b - a$ on $[0, 1]$, $P(\{b\}) = \lim_n P((b - \frac{1}{n}, b]) = 0$: single points have probability zero.

1.2 Conditional probability and independence

Definition (Conditional probability)

For $P(A) > 0$, $P(B \mid A) = P(A \cap B)/P(A)$. As a function of B it is again a probability (on the reduced sample space A).

Theorem 1.3

(a) (Multiplication) $P(A \cap B) = P(B \mid A)P(A)$.

(b) (Total probability) If $A_1, A_2, \ldots$ partition S with $P(A_i) > 0$, then $P(B) = \sum_i P(B \mid A_i)P(A_i)$.

(c) (Bayes) In the same setting, with $P(B) > 0$,

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)} \propto P(B \mid A_j)P(A_j).$$

Proof. (a) is the definition. (b) $B = \bigcup_i (B \cap A_i)$ disjointly; use (A3) and (a). (c) $P(A_j \mid B) = P(B \cap A_j)/P(B)$; expand the numerator by (a) and the denominator by (b). ∎

$P(A_j)$ are the **prior** and $P(A_j \mid B)$ the **posterior** probabilities; Bayes' theorem converts the first into the second after observing B.

Example 1.1

A disease has prevalence 1%. A test detects it with probability 0.95 and gives a false positive with probability 0.05. Then

$$P(\text{disease} \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0095}{0.059} \approx 0.16.$$

A positive result raises the probability from 1% only to about 16%, because healthy people vastly outnumber sick ones.

Definition (Independence)

A, B are independent if $P(A \cap B) = P(A)P(B)$. Events $A_1, \ldots, A_n$ are (mutually) independent if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for *every* subcollection.

If A, B are independent then so are A, B^c: $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)P(B^c)$. Pairwise independence does not imply mutual independence.

Example 1.2

Toss two fair coins: $A = \{\text{first is H}\}$, $B = \{\text{second is H}\}$, $C = \{\text{the two agree}\}$. Each has probability $\frac{1}{2}$ and each pair intersects with probability $\frac{1}{4}$, so the events are pairwise independent; but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$.

1.3 Random variables and distributions

A **random variable** X is a real-valued function on S; events such as $\{X \le x\}$ inherit probabilities.

- **Discrete** X takes values $x_1, x_2, \ldots$; its probability mass function (pmf) $f(x) = P(X = x)$ satisfies $f \ge 0$, $\sum_x f(x) = 1$, $P(X \in A) = \sum_{x \in A} f(x)$.
- **Continuous** X has a probability density function (pdf) $f \ge 0$ with $\int f = 1$ and $P(a \le X \le b) = \int_a^b f(x)\,dx$. Then $P(X = a) = 0$, and $P(a \le X \le a + \Delta) \approx f(a)\Delta$; $f(a)$ itself is not a probability.

We write $X \sim f$. The indicator $I_A(x) = I(x \in A)$ gives compact formulas such as $f(x) = 2x\, I_{[0,1]}(x)$.

1.4 Expectation, variance and moments

Definition (Expectation)

$E[g(X)] = \sum_x g(x)f(x)$ or $\int g(x)f(x)\,dx$, defined when the sum or integral converges absolutely. The mean is $\mu = E(X)$, the variance $\operatorname{Var}(X) = E[(X-\mu)^2]$, the standard deviation $\operatorname{sd}(X) = \sqrt{\operatorname{Var} X}$.

Theorem 1.4

(a) $E(aX+b) = aE(X)+b$ and $E[c_1g_1(X)+c_2g_2(X)] = c_1Eg_1(X)+c_2Eg_2(X)$. (b) If $g_1 \le g_2$ then $Eg_1(X) \le Eg_2(X)$. (c) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$ and $\operatorname{Var} X = E(X^2) - (EX)^2$.

Proof. (a), (b) are linearity and monotonicity of sums and integrals. (c) $\operatorname{Var}(aX+b) = E[a^2(X-\mu)^2]$; and $E(X-\mu)^2 = E(X^2) - 2\mu EX + \mu^2 = E(X^2) - \mu^2$. ■

The **standardized** variable $Z = (X-\mu)/\sigma$ has $EZ = 0$, $\operatorname{Var} Z = 1$. Means need not exist: for $f(k) = \frac{1}{k(k+1)}$, $k = 1, 2, \ldots$, $\sum kf(k) = \sum \frac{1}{k+1} = \infty$.

Example 1.3

For $f(x) = 2x$ on $[0,1]$: $EX = \int_0^1 2x^2\,dx = \frac{2}{3}$, $EX^2 = \int_0^1 2x^3\,dx = \frac{1}{2}$, so $\operatorname{Var} X = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$.

1.5 Distribution functions and generating functions

The **cumulative distribution function** (cdf) is $F(x) = P(X \le x)$. For a discrete X it is a step function with jumps $f(x_k)$ at the x_k; for a continuous X, $F(x) = \int_{-\infty}^x f$ and $F' = f$ at continuity points of f.

Theorem 1.5

(a) F is nondecreasing; (b) $F(-\infty) = 0$, $F(+\infty) = 1$; (c) F is right-continuous; (d) $F(a-) = P(X < a)$, so $F(a) - F(a-) = P(X = a)$.

Proof. (a) $\{X \le x_1\} \subseteq \{X \le x_2\}$. (b), (c), (d) follow from Theorem 1.2 with $\{X \le -k\} \downarrow \emptyset$, $\{X \le k\} \uparrow S$, $\{X \le x + \frac{1}{n}\} \downarrow \{X \le x\}$, $\{X \le a - \frac{1}{n}\} \uparrow \{X < a\}$. ■

A cdf can mix both types, e.g. a waiting time that equals a constant with probability $\frac{1}{2}$ and is uniform otherwise has a jump plus a continuous part.

Probability generating function. For X with values in $\{0, 1, 2, \ldots\}$, $G(s) = E(s^X) = \sum_x s^x f(x)$, $|s| < 1$. It determines the distribution: $f(k) = G^{(k)}(0)/k!$.

Moment generating function. If $E(e^{tX}) < \infty$ for all t in an open interval $(-h, h)$, the mgf is $M(t) = E(e^{tX})$. The kth moment is $m_k = E(X^k)$.

Theorem 1.6

If M exists on $(-h, h)$, then (a) all moments exist, $E(X^k) = M^{(k)}(0)$ and $M(t) = \sum_k \frac{E(X^k)}{k!} t^k$ near 0; (b) if $M_X = M_Y$ on a neighborhood of 0, then X and Y have the same distribution.

Proof of (a). Fix $|t| < h$. Since $|tx|^k/k! \le e^{|tx|} \le e^{tx}+e^{-tx}$, we get $|t|^k E|X|^k/k! \le M(t)+M(-t) < \infty$, so all moments exist. Moreover the partial sums of $\sum_k (tX)^k/k!$ are dominated by $e^{|tX|}$, which is integrable. By dominated convergence $M(t) = E\sum_k \frac{(tX)^k}{k!} = \sum_k \frac{E(X^k)}{k!}t^k$. A power series may be differentiated term by term inside its interval of convergence, so $M^{(k)}(0) = E(X^k)$. Part (b) (uniqueness) is proved with characteristic functions and is beyond these notes. ■

All moments can exist without an mgf (the lognormal distribution).

Example 1.4

For the geometric pmf $f(k) = p(1-p)^{k-1}$, $k = 1, 2, \ldots$: $M(t) = \sum_k e^{tk}p(1-p)^{k-1} = \dfrac{pe^t}{1-(1-p)e^t}$ for $(1-p)e^t < 1$. Then $M'(0) = 1/p$ is the mean.

Cumulants. $C(t) = \log M(t) = \sum_{r\ge 1} c_r \frac{t^r}{r!}$ defines the cumulants c_r. Expanding $\log(1+A) = A - A^2/2 + \ldots$ with $A = m_1 t + m_2 t^2/2 + \ldots$ gives $c_1 = m_1 = \mu$ and $c_2 = m_2 - m_1^2 = \sigma^2$. For the standard exponential, $M(t) = (1-t)^{-1}$, $C(t) = -\log(1-t)$, and $c_r = (r-1)!$.

1.6 Inequalities

Theorem 1.7 (Jensen)

If ϕ is convex on an interval I containing the values of X and EX exists, then $\phi(EX) \le E\phi(X)$.

Proof. A convex function lies above a supporting line at $\mu = EX$: $\phi(x) \ge \phi(\mu) + c(x-\mu)$ for some c and all $x \in I$ (for twice differentiable ϕ, take $c = \phi'(\mu)$ and use Taylor with $\phi'' \ge 0$). Take expectations. ■

E.g. $(EX)^2 \le E(X^2)$, and $E\log X \le \log EX$ for $X > 0$.

Theorem 1.8 (Lyapunov)

If $E|X|^s < \infty$ and $0 < r < s$, then $(E|X|^r)^{1/r} \le (E|X|^s)^{1/s}$.

Proof. $\phi(y) = y^{s/r}$ is convex on $[0,\infty)$. With $Y = |X|^r$ (integrable since $|X|^r \le 1 + |X|^s$), Jensen gives $(EY)^{s/r} \le E(Y^{s/r}) = E|X|^s$. ■

Theorem 1.9 (Markov and Chebyshev)

(a) If $E|Z|^r < \infty$ $(r > 0)$, then $P(|Z| \ge k) \le E|Z|^r/k^r$ for $k > 0$. (b) If $\operatorname{Var} X < \infty$, then $P(|X - EX| \ge k) \le \operatorname{Var} X/k^2$.

Proof. (a) $I(|Z| \ge k) \le (|Z|/k)^r$ pointwise; take expectations. (b) Apply (a) to $Z = X - EX$ with $r = 2$. ■

So $P(|X-\mu| \ge 3\sigma) \le \frac{1}{9}$ for *every* distribution with finite variance (for the normal the exact value is 0.0027).

Theorem 1.10

If $\operatorname{Var} X = 0$ then $P(X = \mu) = 1$.

Proof. $\{|X-\mu|>0\} = \bigcup_n \{|X-\mu| \geq \frac{1}{n}\}$, and each set has probability $\leq \operatorname{Var} X \cdot n^2 = 0$ by Chebyshev; countable subadditivity finishes. ■

Chapter 2. Multivariate Random Variables

2.1 Joint and marginal distributions

A pair (X, Y) is described by a **joint pmf** $f(x, y) = P(X = x, Y = y)$ in the discrete case, or a **joint pdf** $f \geq 0$ with $P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$ in the continuous case. In both cases $f \geq 0$ and f sums or integrates to 1.

Definition (Marginal distributions)

$f_1(x) = \sum_y f(x, y)$ or $\int f(x, y)\, dy$, and $f_2(y) = \sum_x f(x, y)$ or $\int f(x, y)\, dx$.

Marginals are obtained by "summing out" the other variable; in a table they are the row and column totals. The joint distribution determines the marginals, but not conversely.

The **joint cdf** is $F(x, y) = P(X \leq x, Y \leq y)$. For continuous (X, Y), $f = \partial^2 F / \partial x \partial y$ at continuity points. The marginal cdfs are limits: $F_1(x) = \lim_{y \to \infty} F(x, y)$.

Theorem 2.1

(a) F is nondecreasing in each argument, and for $x_1 < x_2$, $y_1 < y_2$,

$$P(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1) \geq 0.$$

(b) $F(x, y) \to 0$ as $x \to -\infty$ or $y \to -\infty$, and $F \to 1$ as both $\to +\infty$. (c) F is right-continuous in each argument.

Proof. The rectangle identity is inclusion–exclusion for $\{X \leq x_2, Y \leq y_2\}$ minus the two strips. (b), (c) follow from continuity of probability (Theorem 1.2) as in one dimension. ∎

Example 2.1

Let $f(x, y) = c(x + y)$ on $0 \leq x, y \leq 1$. Then $\int_0^1 \int_0^1 (x + y)\, dy\, dx = 1$, so $c = 1$. The marginal is $f_1(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}$ on $[0, 1]$, and by symmetry $f_2(y) = y + \frac{1}{2}$. Also $P(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y)\, dy\, dx = \int_0^1 \frac{1 - x^2}{2}\, dx = \frac{1}{3}$.

2.2 Expectations, covariance and correlation

$E[g(X, Y)] = \sum\sum g(x, y) f(x, y)$ or $\iint g f$; it is linear and monotone. If g depends only on x, $Eg(X)$ can be computed from the joint or from the marginal: integrating out y first gives $\int g(x) f_1(x)\, dx$.

Definition (Covariance and correlation)

$\operatorname{Cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)]$ and, for positive variances, $\rho = \operatorname{Corr}(X, Y) = \operatorname{Cov}(X, Y)/(\sigma_1 \sigma_2)$.

Theorem 2.2

(a) $\operatorname{Cov}(X, Y) = \operatorname{Cov}(Y, X)$ and $\operatorname{Cov}(X, X) = \operatorname{Var} X$. (b) $\operatorname{Cov}(aX + b, cY + d) = ac \operatorname{Cov}(X, Y)$. (c) $\operatorname{Cov}(X, Y) = E(XY) - E(X)E(Y)$.

Proof. (a) is immediate. (b) $aX+b-E(aX+b)=a(X-\mu_1)$, and similarly for Y. (c) Expand $(X-\mu_1)(Y-\mu_2)$ and use linearity. ■

Theorem 2.3

(a) $\operatorname{Var}\left(\frac{Y-\mu_2}{\sigma_2}-\rho\frac{X-\mu_1}{\sigma_1}\right)=1-\rho^2$. (b) $-1\le\rho\le 1$. (c) $\rho=\pm1$ iff $\frac{Y-\mu_2}{\sigma_2}=\pm\frac{X-\mu_1}{\sigma_1}$ with probability 1.

Proof. (a) Expand: $1-2\rho\cdot\rho+\rho^2=1-\rho^2$. (b) A variance is nonnegative. (c) $\rho=\pm1$ makes the variance in (a) zero, so the variable is constant (zero, since its mean is zero) with probability 1 (Theorem 1.10). ■

Equivalently,

$$|\operatorname{Cov}(X,Y)|\le\sigma_1\sigma_2,$$

a case of the **Cauchy–Schwarz inequality**, $|E(VW)|\le\sqrt{EV^2\,EW^2}$. The correlation measures *linear* association, free of units.

Example 2.2 (Example 2.1 continued)

$EX=\int_0^1 x(x+\frac12)\,dx=\frac{7}{12}$, $EX^2=\int_0^1 x^2(x+\frac12)\,dx=\frac{5}{12}$, so $\operatorname{Var}X=\frac{5}{12}-\frac{49}{144}=\frac{11}{144}$. Also $E(XY)=\int_0^1\int_0^1 xy(x+y)\,dy\,dx=\frac13$, so $\operatorname{Cov}(X,Y)=\frac13-\frac{49}{144}=-\frac{1}{144}$ and $\rho=-\frac{1}{11}$.

Joint moments and generating functions. The (r,s) joint moment is $E(X^rY^s)$. If $M(t_1,t_2)=E(e^{t_1X+t_2Y})$ is finite near $(0,0)$, then (a) all joint moments exist and $E(X^rY^s)=\partial^{r+s}M/\partial t_1^r\partial t_2^s$ at 0, and (b) M determines the joint distribution. The marginal mgfs are $M_X(s)=M(s,0)$ and $M_Y(t)=M(0,t)$. The joint cumulant generating function $C=\log M$ has $\partial C/\partial t_i(0)$ equal to the means and $\partial^2C/\partial t_i\partial t_j(0)$ equal to the variances and covariance.

2.3 Conditional distributions and conditional expectation

Definition (Conditional pmf/pdf)

For $f_1(x)>0$, $f_{2|1}(y\mid x)=f(x,y)/f_1(x)$. For each fixed x it is a pmf/pdf in y.

In the continuous case $P(X=x)=0$, so this is justified as a limit: if f and f_1 are continuous,

$$P(c\le Y\le d\mid x\le X\le x+h)=\frac{\frac1h\int_x^{x+h}\int_c^d f(s,y)\,dy\,ds}{\frac1h\int_x^{x+h}f_1(s)\,ds}\xrightarrow{h\downarrow0}\int_c^d f_{2|1}(y\mid x)\,dy.$$

Theorem 2.4

$P(a\le X\le b,c\le Y\le d)=\int_a^b P(c\le Y\le d\mid X=x)f_1(x)\,dx$ (a sum in the discrete case).

Proof. Substitute $f(x,y)=f_{2|1}(y\mid x)f_1(x)$ into the double integral and integrate in y first. ■

Definition (Conditional expectation and variance)

$E[g(X,Y)\mid X=x]=\int g(x,y)f_{2|1}(y\mid x)\,dy$; $\mu_{2|1}(x)=E(Y\mid X=x)$; $\operatorname{Var}(Y\mid X=x)=E[(Y-\mu_{2|1}(x))^2\mid X=x]=E(Y^2\mid X=x)-\mu_{2|1}(x)^2$. Replacing x by X gives the random variables $E(Y\mid X)$ and $\operatorname{Var}(Y\mid X)$.

Conditional expectation is linear and monotone, and functions of the conditioning variable factor out: $E[c(X)g(Y) \mid X = x] = c(x)E[g(Y) \mid X = x]$.

Theorem 2.5

(a) (Tower property) $E[E(Y \mid X)] = E(Y)$. (b) $\operatorname{Cov}(Y - E(Y \mid X), v(X)) = 0$ for every function v with $Ev(X)^2 < \infty$.

Proof. (a) $\int \left(\int y f_{2|1}(y \mid x)\, dy \right) f_1(x)\, dx = \iint y f(x,y)\, dy\, dx = EY$. (b) Let $Z = Y - E(Y \mid X)$. Then $E(Z \mid X) = 0$, so $EZ = 0$ by (a), and $E[Zv(X)] = E[v(X)E(Z \mid X)] = 0$. ■

Theorem 2.6 (Best predictor)

Among all functions u, $E[(Y - u(X))^2]$ is minimized by $u(X) = E(Y \mid X)$, and the minimum is $E[\operatorname{Var}(Y \mid X)]$.

Proof. Write $Y - u(X) = (Y - E(Y \mid X)) + v(X)$ with $v = E(Y \mid X) - u(X)$. The cross term has expectation 0 by Theorem 2.5(b), so $E(Y-u)^2 = E(Y - E(Y \mid X))^2 + Ev(X)^2$, minimized when $v = 0$. And $E(Y - E(Y \mid X))^2 = E[E((Y - E(Y \mid X))^2 \mid X)] = E \operatorname{Var}(Y \mid X)$. ■

$E(Y \mid X)$ is called the **regression function** of Y on X.

Theorem 2.7 (Variance decomposition)

$\operatorname{Var} Y = E[\operatorname{Var}(Y \mid X)] + \operatorname{Var}[E(Y \mid X)]$.

Proof. $Y - \mu = (Y - E(Y \mid X)) + (E(Y \mid X) - \mu)$; the two parts are uncorrelated by Theorem 2.5(b). Square and take expectations, using $E[E(Y \mid X)] = \mu$ and Theorem 2.6. ■

Geometrically $Y - EY$ splits into orthogonal pieces: the part explained by X and the residual. This is the basis of regression and ANOVA (Chapter 10).

Example 2.3 (Example 2.1 continued)

$f_{2|1}(y \mid x) = \frac{x+y}{x+1/2}$ on $[0,1]$, so $E(Y \mid X = x) = \frac{x/2+1/3}{x+1/2} = \frac{3x+2}{6x+3}$. The regression is not linear, and it decreases in x, consistent with $\rho < 0$.

2.4 Independence

Definition (Independent random variables)

X and Y are independent if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all (Borel) sets A, B.

Theorem 2.8

Each of the following is equivalent to independence: (a) $F(x,y) = F_1(x)F_2(y)$ for all x, y; (b) $f(x,y) = f_1(x)f_2(y)$ for all x, y; (c) $M(t_1,t_2) = M_X(t_1)M_Y(t_2)$ near 0 (when the mgf exists). Moreover (b) holds iff $f(x,y) = g_1(x)g_2(y)$ for *some* functions g_1, g_2.

Proof. Rectangles determine the distribution, so (a)$\Leftrightarrow$ independence. (a) and (b) are equivalent by differentiation and integration. (c) follows from the uniqueness of the mgf. If $f = g_1 g_2$, integrating gives $f_1 = g_1 \int g_2$, $f_2 = g_2 \int g_1$ and $1 = \int g_1 \int g_2$, so $f = f_1 f_2$. ■

The factorization must hold on the whole plane, *support included*: $f(x,y) = 8xy$ on $0 \le x \le y \le 1$ looks like a product, but the triangular support makes X, Y dependent.

Theorem 2.9

If X, Y are independent then (a) $g_1(X)$ and $g_2(Y)$ are independent; (b) $E[g_1(X)g_2(Y)] = Eg_1(X)\,Eg_2(Y)$; (c) $\operatorname{Cov}(X,Y) = 0$; (d) $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y$. In general $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X,Y)$.

Proof. (a) $\{g_1(X) \in A\} = \{X \in g_1^{-1}(A)\}$. (b) Factor the double integral. (c) is (b) with identities and Theorem 2.2(c). (d) Expand the square. ∎

Uncorrelated does not imply independent: if X is uniform on $\{-1,0,1\}$ and $Y = X^2$, then $\operatorname{Cov}(X,Y) = EX^3 = 0$, yet Y is a function of X.

2.5 Random vectors

For $\mathbf{X} = (X_1, \dots, X_k)'$ the joint pmf/pdf, marginals (sum or integrate out the other coordinates), cdf and expectations are defined as above. The **mean vector** is $E\mathbf{X} = (EX_1, \dots, EX_k)'$, and the **covariance matrix** is

$$\operatorname{Var}(\mathbf{X}) = E[(\mathbf{X}-\mu)(\mathbf{X}-\mu)'] = \big(\operatorname{Cov}(X_i, X_j)\big)_{i,j}.$$

More generally $\operatorname{Cov}(\mathbf{X}, \mathbf{Y}) = E[(\mathbf{X}-\mu)(\mathbf{Y}-\eta)']$.

Theorem 2.10

For constant matrices A, C and vectors $\mathbf{b}, \mathbf{d}$:

(a) $E(A\mathbf{X} + \mathbf{b}) = AE\mathbf{X} + \mathbf{b}$;

(b) $\operatorname{Var}(A\mathbf{X}+\mathbf{b}) = A\operatorname{Var}(\mathbf{X})A'$ and $\operatorname{Cov}(A\mathbf{X}+\mathbf{b}, C\mathbf{Y}+\mathbf{d}) = A\operatorname{Cov}(\mathbf{X},\mathbf{Y})C'$;

(c) $\operatorname{Var}(\mathbf{X}+\mathbf{Y}) = \operatorname{Var}\mathbf{X} + \operatorname{Var}\mathbf{Y} + \operatorname{Cov}(\mathbf{X},\mathbf{Y}) + \operatorname{Cov}(\mathbf{Y},\mathbf{X})$;

(d) $\operatorname{Var}(\mathbf{X})$ is symmetric and nonnegative definite.

Proof. (a)–(c) follow from $E(CWD) = CE(W)D$ for a random matrix W, applied to $(\mathbf{X}-\mu)(\mathbf{Y}-\eta)'$. (d) $\mathbf{a}'\operatorname{Var}(\mathbf{X})\mathbf{a} = \operatorname{Var}(\mathbf{a}'\mathbf{X}) \ge 0$. ∎

By the spectral theorem $\operatorname{Var}\mathbf{X} = P\operatorname{diag}(\lambda_1, \dots, \lambda_k)P'$ with P orthogonal and $\lambda_1 \ge \cdots \ge \lambda_k \ge 0$. Then $P'\mathbf{X}$ has uncorrelated coordinates with variances λ_i: these are the **principal components**, and the eigenvectors give the main directions of spread.

Generating functions. $M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}})$ yields all joint moments and determines the distribution. The cgf $C = \log M$ gives $E\mathbf{X} = \nabla C(\mathbf{0})$ and $\operatorname{Var}\mathbf{X} = \nabla^2 C(\mathbf{0})$ (gradient and Hessian).

Conditioning. $f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y} \mid \mathbf{x}) = f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})/f_{\mathbf{X}}(\mathbf{x})$. The tower property, the orthogonality $\operatorname{Cov}(\mathbf{Y} - E(\mathbf{Y} \mid \mathbf{X}), v(\mathbf{X})) = 0$, the matrix decomposition

$$\operatorname{Var}\mathbf{Y} = E[\operatorname{Var}(\mathbf{Y} \mid \mathbf{X})] + \operatorname{Var}[E(\mathbf{Y} \mid \mathbf{X})],$$

and the best-predictor property for $E\|\mathbf{Y} - u(\mathbf{X})\|^2$ carry over with the same proofs.

Independence. $X_1, \ldots, X_n$ are independent iff the joint pdf (or the mgf) factors into marginals, again iff it factors into *some* functions of the separate arguments. Then functions of separate components are independent, expectations of products factor, the covariance matrix of $\sum_i \mathbf{X}_i$ is $\sum_i \operatorname{Var} \mathbf{X}_i$, and $M_{\sum \mathbf{X}_i}(\mathbf{t}) = \prod_i M_{\mathbf{X}_i}(\mathbf{t})$.

Example 2.4

Let X_1, X_2, X_3 be i.i.d. standard exponential and $Y_1 = X_1$, $Y_2 = X_1 + X_2$. Then $\operatorname{Var}(Y_1, Y_2)' = A\, I\, A'$ with $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, i.e. $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, so $\operatorname{Corr}(Y_1, Y_2) = 1/\sqrt{2}$.

Chapter 3. Common Distributions

Conventions

Many distributions below are introduced through a **representation**: $X \overset{d}{=} Z_1 + \cdots + Z_n$ means X has the same distribution as a sum of the indicated variables. Representations make means, variances and mgfs almost free, by Theorem 2.9 and the product rule for mgfs of independent sums. Throughout, $\mathrm{Exp}(\beta)$ and $\mathrm{Gamma}(\alpha, \beta)$ use the *scale* β (mean β and $\alpha\beta$).

3.1 The hypergeometric distribution

Draw n items without replacement from N items of which D are "marked". The number X of marked items drawn has

$$f(x) = \binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n}, \qquad X \sim H(n; N, D).$$

Proposition 3.1

With $p = D/N$: $EX = np$ and $\operatorname{Var} X = \dfrac{N-n}{N-1}\, np(1-p)$.

Proof. Write $X = \sum_{i=1}^{n} I_i$, where I_i indicates that the ith draw is marked. By symmetry $P(I_i = 1) = p$, and for $i \neq j$, $P(I_i = I_j = 1) = \frac{D(D-1)}{N(N-1)}$. So $EX = np$ and

$$\operatorname{Var} X = np(1-p) + n(n-1)\Big(\frac{D(D-1)}{N(N-1)} - p^2\Big) = np(1-p)\Big(1 - \frac{n-1}{N-1}\Big).$$

■

The factor $\frac{N-n}{N-1}$ is the finite-population correction (cf. sampling without replacement). As $N \to \infty$ with $D/N \to p$ and n fixed, $f(x) \to \binom{n}{x}p^x(1-p)^{n-x}$: sampling without replacement from a huge population is like sampling with replacement.

3.2 Binomial and multinomial distributions

Definition (Binomial)

$X \sim B(n, p)$ if $f(x) = \binom{n}{x}p^x(1-p)^{n-x}$, $x = 0, \ldots, n$. Equivalently $X \overset{d}{=} Z_1 + \cdots + Z_n$ with $Z_i \overset{\text{i.i.d.}}{\sim} \mathrm{Bernoulli}(p)$.

Proposition 3.2

$M(t) = (1 - p + pe^t)^n$, $EX = np$, $\operatorname{Var} X = np(1-p)$. If $X_i \sim B(n_i, p)$ are independent then $\sum X_i \sim B(\sum n_i, p)$.

Proof. Each Bernoulli has mgf $1 - p + pe^t$, mean p and variance $p(1-p)$; multiply mgfs and add means and variances. For the sum, the mgfs multiply to $(1 - p + pe^t)^{\sum n_i}$, and the mgf determines the distribution. ■

Definition (Multinomial)

Each of n independent trials falls in category j with probability p_j ($\sum_j p_j = 1$). The counts $\mathbf{X} = (X_1, \dots, X_k)$ have

$$f(\mathbf{x}) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}, \qquad \sum x_j = n, \qquad \mathbf{X} \sim \text{Multi}(n, \mathbf{p}).$$

Equivalently $\mathbf{X} \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with i.i.d. $\mathbf{Z}_i \sim \text{Multi}(1, \mathbf{p})$ (one-hot vectors).

Proposition 3.3

$X_j \sim B(n, p_j)$; $E\mathbf{X} = n\mathbf{p}$; $\operatorname{Var} \mathbf{X} = n(\operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}')$, i.e. $\operatorname{Cov}(X_i, X_j) = -np_ip_j$ for $i \neq j$. Merging categories gives again a multinomial.

Proof. For one trial, $\mathbf{Z}$ is one-hot with $E\mathbf{Z} = \mathbf{p}$ and $E(Z_iZ_j) = 0$ for $i \neq j$, so $\operatorname{Var} \mathbf{Z} = \operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}'$. Sum over n i.i.d. trials. A merged count is a sum of indicators of a union of categories. ■

The negative covariances reflect the constraint $\sum X_j = n$.

3.3 Geometric and negative binomial distributions

Definition (Geometric and negative binomial)

In i.i.d. Bernoulli(p) trials, the trial number X of the first success is Geo(p): $f(x) = (1-p)^{x-1}p$, $x = 1, 2, \dots$. The trial number of the rth success is Negbin(r, p): $f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$, $x = r, r+1, \dots$. Equivalently $\text{Negbin}(r, p) \overset{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \overset{\text{i.i.d.}}{\sim} \text{Geo}(p)$.

Proposition 3.4

For $X \sim \text{Geo}(p)$: $M(t) = \frac{pe^t}{1-(1-p)e^t}$ for $(1-p)e^t < 1$, $EX = \frac{1}{p}$, $\operatorname{Var} X = \frac{1-p}{p^2}$. Hence Negbin($r, p$) has mean $\frac{r}{p}$ and variance $\frac{r(1-p)}{p^2}$. Moreover the geometric distribution is **memoryless**: $P(X > m + k \mid X > m) = P(X > k)$.

Proof. The mgf is a geometric series. From $C(t) = \log M(t)$, $C'(0) = 1 + \frac{1-p}{p} = \frac{1}{p}$, and differentiating once more gives $\frac{1-p}{p^2}$. For memorylessness, $P(X > k) = (1-p)^k$, so the ratio is $(1-p)^{m+k}/(1-p)^m$. The waiting times between successive successes are i.i.d. geometric, which gives the representation. ■

3.4 The Poisson distribution

Definition (Poisson)

$X \sim \text{Poisson}(\lambda)$ if $f(x) = e^{-\lambda}\lambda^x/x!$, $x = 0, 1, \dots$.

Proposition 3.5 (Poisson approximation)

If $np_n \to \lambda > 0$, then $\binom{n}{x} p_n^x (1-p_n)^{n-x} \to e^{-\lambda}\lambda^x/x!$ for every x.

Proof. $\binom{n}{x}p_n^x = \frac{n(n-1)\cdots(n-x+1)}{n^x}\cdot\frac{(np_n)^x}{x!} \to \frac{\lambda^x}{x!}$, and $(1-p_n)^{n-x} = (1-\frac{np_n}{n})^n(1-p_n)^{-x} \to e^{-\lambda}$, using $(1+a_n/n)^n \to e^a$ when $a_n \to a$. ■

So for many trials with a small success probability, the number of successes is approximately Poisson. E.g. $B(500, 0.004)$ gives $P(X=0) = 0.996^{500} = 0.135$, against $e^{-2} = 0.135$.

Proposition 3.6

$M(t) = \exp\{\lambda(e^t-1)\}$; $EX = \operatorname{Var} X = \lambda$; independent Poisson($\lambda_i$) variables sum to Poisson($\sum\lambda_i$).

Proof. $\sum_x e^{tx}e^{-\lambda}\lambda^x/x! = e^{-\lambda}e^{\lambda e^t}$. The cgf $\lambda(e^t-1)$ has all derivatives at 0 equal to λ. Mgfs of independent summands multiply. ■

The Poisson process. Let N_t count events in $[0,t]$. Assume (a) stationarity: $N_{s+t} - N_s \stackrel{d}{=} N_t$, and $N_0 = 0$; (b) independent increments over disjoint intervals; (c) $P(N_h = 1) = \lambda h + o(h)$; (d) $P(N_h \geq 2) = o(h)$ as $h \downarrow 0$.

Theorem 3.7

Under (a)–(d), $N_t \sim$ Poisson(λt).

Proof. Let $g(x,t) = P(N_t = x)$. Decompose by the increment on $(t, t+h]$ and use (a), (b):

$$g(x,t+h) = g(x,t)P(N_h=0) + g(x-1,t)P(N_h=1) + o(h) = g(x,t)(1-\lambda h) + g(x-1,t)\lambda h + o(h).$$

Hence $\partial_t g(x,t) = -\lambda g(x,t) + \lambda g(x-1,t)$, i.e. $\partial_t\{e^{\lambda t}g(x,t)\} = \lambda e^{\lambda t}g(x-1,t)$, with $g(0,0) = 1$ and $g(x,0) = 0$ for $x \geq 1$. Solving successively for $x = 0, 1, 2, \ldots$ gives $g(0,t) = e^{-\lambda t}$, then $g(x,t) = e^{-\lambda t}(\lambda t)^x/x!$ by induction. ■

Heuristically: split $[0,t]$ into n pieces; each has an event with probability about $\lambda t/n$, independently, so $N_t \approx B(n, \lambda t/n) \to$ Poisson(λt).

3.5 Exponential and gamma distributions

In a Poisson process with rate λ, the waiting time W_1 to the first event satisfies $P(W_1 > t) = P(N_t = 0) = e^{-\lambda t}$, so it has density $\lambda e^{-\lambda t}$: the **exponential distribution** Exp(β) with mean $\beta = 1/\lambda$. Like the geometric, it is memoryless: $P(W > s+t \mid W > s) = P(W > t)$.

The waiting time W_r to the rth event satisfies $P(W_r > t) = P(N_t \leq r-1)$; differentiating gives the density $\frac{\lambda^r}{(r-1)!}t^{r-1}e^{-\lambda t}$. Allowing a real shape $\alpha > 0$:

Definition (Gamma)

$X \sim$ Gamma(α, β) if $f(x) = \dfrac{1}{\Gamma(\alpha)\beta^\alpha}x^{\alpha-1}e^{-x/\beta}$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}\,dx$; α is the shape and β the scale. Exp(β) = Gamma($1, \beta$).

Recall $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$, $\Gamma(n) = (n-1)!$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Proposition 3.8

For $X \sim$ Gamma(α, β): $EX = \alpha\beta$, $\operatorname{Var} X = \alpha\beta^2$, $M(t) = (1-\beta t)^{-\alpha}$ for $t < 1/\beta$. Independent Gamma(α_i, β) variables (common scale) sum to Gamma($\sum\alpha_i, \beta$). In particular Gamma(r, β) $\stackrel{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \stackrel{\text{i.i.d.}}{\sim}$ Exp(β), which identifies W_r above.

Proof. Substitute $y = x/\beta$: $EX^k = \beta^k\Gamma(\alpha+k)/\Gamma(\alpha)$, giving $\alpha\beta$ and $\alpha(\alpha+1)\beta^2$. Substitute $y = (1/\beta - t)x$ in $\int e^{tx} f(x)\,dx$ to get $(1-\beta t)^{-\alpha}$. Multiply the mgfs for the sum. ■

3.6 The normal distribution

Definition (Normal)

$Z \sim N(0,1)$ has density $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and cdf Φ. $X \sim N(\mu, \sigma^2)$ if $X \overset{d}{=} \mu + \sigma Z$, i.e. $f(x) = \frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$.

That $\int \phi = 1$ follows from $(\int e^{-z^2/2} dz)^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = 2\pi$ in polar coordinates.

Proposition 3.9

(a) $EX = \mu$, $\operatorname{Var} X = \sigma^2$. (b) $M(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$. (c) $aX + b \sim N(a\mu + b, a^2\sigma^2)$, and independent normals sum to a normal with the means and variances added.

Proof. (b) $Ee^{tZ} = \int \frac{1}{\sqrt{2\pi}} e^{-(z-t)^2/2} e^{t^2/2}\,dz = e^{t^2/2}$ (complete the square); then $Ee^{t(\mu+\sigma Z)} = e^{\mu t} e^{\sigma^2 t^2/2}$. (a) The cgf $\mu t + \frac{1}{2}\sigma^2 t^2$ has first and second derivatives μ and σ^2 at 0. (c) Compare mgfs. ■

Quantiles. The upper α point z_α satisfies $P(Z > z_\alpha) = \alpha$ (e.g. $z_{0.05} = 1.645$, $z_{0.025} = 1.96$), and the upper α point of $N(\mu, \sigma^2)$ is $\mu + \sigma z_\alpha$. Probabilities are computed by standardizing: $P(X \le x) = \Phi((x-\mu)/\sigma)$.

The normal arises as the limit of standardized binomials (de Moivre–Laplace), $P(a \le \frac{X-np}{\sqrt{np(1-p)}} \le b) \to \Phi(b) - \Phi(a)$, a special case of the central limit theorem (Chapter 5).

Summary of representations

distribution	pmf/pdf	representation
$B(n,p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	sum of n i.i.d. Bernoulli(p)
Multi($n, \mathbf{p}$)	$\frac{n!}{\prod x_j!} \prod p_j^{x_j}$	sum of n i.i.d. Multi($1, \mathbf{p}$)
Negbin(r,p)	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$	sum of r i.i.d. Geo(p)
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	limit of $B(n, \lambda/n)$
Gamma(r, β)	$\frac{x^{r-1}e^{-x/\beta}}{\Gamma(r)\beta^r}$	sum of r i.i.d. Exp(β)
$N(\mu, \sigma^2)$	$\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$	$\mu + \sigma Z$, $Z \sim N(0,1)$

Chapter 4. Sampling Distributions

4.1 Distributions of statistics

A **random sample** $X_1, \dots, X_n$ is i.i.d. from a population pdf f; a **statistic** is a function of the sample not involving unknown parameters, and its distribution is its **sampling distribution**. For instance, for a Bernoulli(p) sample, $n\hat{p} = \sum X_i \sim B(n, p)$, so $E\hat{p} = p$ and $\operatorname{Var}\hat{p} = p(1-p)/n$: $\hat{p}$ concentrates around p.

Discrete case. If $Y = u(\mathbf{X})$, then $f_Y(y) = \sum_{\mathbf{x}: u(\mathbf{x})=y} f_{\mathbf{X}}(\mathbf{x})$. Convolution sums reproduce the closure of binomial and Poisson families under addition.

Continuous case: change of variables.

Theorem 4.1

Let $\mathbf{X}$ have pdf $f_{\mathbf{X}}$ supported on an open set $\mathcal{X} \subseteq \mathbb{R}^n$, and let $u : \mathcal{X} \to \mathcal{Y}$ be one-to-one and onto with differentiable inverse $\mathbf{x} = u^{-1}(\mathbf{y})$ and Jacobian $J(\mathbf{y}) = \det(\partial x_i / \partial y_j) \neq 0$. Then $\mathbf{Y} = u(\mathbf{X})$ has pdf

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(u^{-1}(\mathbf{y}))\,|J(\mathbf{y})|, \qquad \mathbf{y} \in \mathcal{Y}.$$

If instead $\mathcal{X}$ splits into pieces $\mathcal{X}_1, \dots, \mathcal{X}_m$ on each of which u is one-to-one onto $\mathcal{Y}$, then $f_{\mathbf{Y}}(\mathbf{y}) = \sum_j f_{\mathbf{X}}(u_j^{-1}(\mathbf{y}))|J_j(\mathbf{y})|$.

Proof. For $B \subseteq \mathcal{Y}$, $P(\mathbf{Y} \in B) = P(\mathbf{X} \in u^{-1}(B)) = \int_{u^{-1}(B)} f_{\mathbf{X}}(\mathbf{x})\,d\mathbf{x} = \int_B f_{\mathbf{X}}(u^{-1}(\mathbf{y}))|J(\mathbf{y})|\,d\mathbf{y}$ by the change-of-variables formula for integrals. In the many-to-one case, add the contributions of the pieces. ■

To get the pdf of a single $Y_1 = u_1(\mathbf{X})$, complete it to a one-to-one map $(Y_1, \dots, Y_n)$ by adding convenient coordinates, apply the theorem, then integrate out the extra coordinates.

Example 4.1

$X \sim N(0,1)$ and $Y = X^2$: the map $x \mapsto x^2$ is two-to-one, with branches $x = \pm\sqrt{y}$ and $|J| = \frac{1}{2\sqrt{y}}$. So $f_Y(y) = 2\phi(\sqrt{y})\frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}$, $y > 0$: the Gamma($\frac{1}{2}, 2$) density.

Example 4.2 (Beta and gamma)

Let $X_1 \sim \text{Gamma}(\alpha_1, 1)$ and $X_2 \sim \text{Gamma}(\alpha_2, 1)$ be independent, $Y_1 = X_1 + X_2$ and $Y_2 = X_1/(X_1 + X_2)$. The inverse is $x_1 = y_1 y_2$, $x_2 = y_1(1 - y_2)$ with $|J| = y_1$, so

$$f(y_1, y_2) = \frac{y_1^{\alpha_1+\alpha_2-1} e^{-y_1}}{\Gamma(\alpha_1+\alpha_2)} \cdot \frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_2^{\alpha_1-1}(1-y_2)^{\alpha_2-1}.$$

Hence $Y_1 \sim \text{Gamma}(\alpha_1 + \alpha_2, 1)$ and $Y_2 \sim \text{Beta}(\alpha_1, \alpha_2)$, and they are independent. As a by-product, $B(\alpha_1, \alpha_2) = \int_0^1 y^{\alpha_1-1}(1-y)^{\alpha_2-1} dy = \Gamma(\alpha_1)\Gamma(\alpha_2)/\Gamma(\alpha_1+\alpha_2)$.

Example 4.3

For $X_1, X_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, the mean $(X_1 + X_2)/2$ has the triangular density $2 - 4|y - \frac{1}{2}|$ on $(0,1)$. (Use $Y = (X_1 + X_2)/2$, $Z = (X_1 - X_2)/2$, $|J| = 2$, and integrate out z.) In general $U(a,b) \overset{d}{=} a + (b-a)U(0,1)$.

Probability integral transform. If F is continuous and $X \sim F$, then $F(X) \sim U(0,1)$. Conversely, if $U \sim U(0,1)$, then $F^{-1}(U) \sim F$ with $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. This is the basis of simulation.

4.2 Chi-square, t and F distributions

Definition (Chi-square)

$\chi^2(r) = \text{Gamma}(r/2, 2)$. For integer r, $\chi^2(r) \overset{d}{=} Z_1^2 + \cdots + Z_r^2$ with $Z_i \overset{\text{i.i.d.}}{\sim} N(0,1)$ (Example 4.1 and the gamma sum rule).

So $E = r$, $\text{Var} = 2r$, $M(t) = (1-2t)^{-r/2}$, and independent chi-squares add their degrees of freedom. The upper α point $\chi^2_\alpha(r)$ satisfies $P(Y > \chi^2_\alpha(r)) = \alpha$. Since $X \sim \text{Gamma}(\nu, \beta) \iff 2X/\beta \sim \chi^2(2\nu)$, gamma quantiles are $\frac{\beta}{2}\chi^2_\alpha(2\nu)$.

Theorem 4.2 (Normal samples)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, $\bar{X} = \frac{1}{n}\sum X_i$, $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$. Then

(a) $\bar{X} \sim N(\mu, \sigma^2/n)$;

(b) $\bar{X}$ and S^2 are independent;

(c) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$.

Proof. Standardize so that $\mu = 0$ and $\sigma = 1$, and choose an orthogonal matrix P whose first row is $(1/\sqrt{n}, \ldots, 1/\sqrt{n})$ and set $\mathbf{Y} = P\mathbf{X}$. Since $\mathbf{X} \sim N_n(\mathbf{0}, I)$, also $\mathbf{Y} \sim N_n(\mathbf{0}, PP') = N_n(\mathbf{0}, I)$ (Section 4.4), so $Y_1, \ldots, Y_n$ are i.i.d. $N(0,1)$. Now $Y_1 = \sqrt{n}\bar{X}$, and $\sum_i Y_i^2 = \sum_i X_i^2$, so

$$\textstyle\sum_{i \geq 2} Y_i^2 = \sum X_i^2 - n\bar{X}^2 = (n-1)S^2.$$

. Hence $\bar{X}$ depends only on Y_1 and S^2 only on $Y_2, \ldots, Y_n$, which gives (b). Also $(n-1)S^2$ is a sum of $n-1$ squared standard normals, giving (c). (a) holds since $Y_1 \sim N(0,1)$. ∎

Definition (Student's t and Snedecor's F)

If $Z \sim N(0,1)$ and $V \sim \chi^2(r)$ are independent, then $T = Z/\sqrt{V/r} \sim t(r)$. If $V_i \sim \chi^2(r_i)$ are independent, then $F = (V_1/r_1)/(V_2/r_2) \sim F(r_1, r_2)$.

Proposition 4.3

$$f_{t(r)}(x) = \frac{\Gamma(\frac{r+1}{2})}{\sqrt{r\pi}\,\Gamma(\frac{r}{2})}\Big(1 + \frac{x^2}{r}\Big)^{-\frac{r+1}{2}},$$

$$f_{F(r_1,r_2)}(x) = \frac{\Gamma(\frac{r_1+r_2}{2})}{\Gamma(\frac{r_1}{2})\Gamma(\frac{r_2}{2})}\Big(\frac{r_1}{r_2}\Big)^{r_1/2}\frac{x^{r_1/2-1}}{(1+\frac{r_1}{r_2}x)^{(r_1+r_2)/2}}.$$

Moreover $T^2 \sim F(1,r)$, $1/F(r_1,r_2) \sim F(r_2,r_1)$, and $t(r) \to N(0,1)$ as $r \to \infty$.

Proof. For t: map $(Z,V) \mapsto (T,V)$ with $z = t\sqrt{v/r}$, $|J| = \sqrt{v/r}$, and integrate v out using the gamma integral. For F: map $(V_1,V_2) \mapsto (F,V_2)$ with $v_1 = r_1 f v_2/r_2$, $|J| = r_1 v_2/r_2$, and integrate v_2 out. $T^2 = Z^2/(V/r)$ with $Z^2 \sim \chi^2(1)$. Finally $V/r \to 1$ (its variance is $2/r \to 0$), so $T \to Z$ in distribution (Chapter 5). ∎

Corollary 4.4 (Inference for a normal mean and variance)

For a normal sample, $\sqrt{n}(\bar{X}-\mu)/S \sim t(n-1)$ and $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$. Hence the $1-\alpha$ confidence intervals

$$\mu \in \bar{X} \pm t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}, \qquad \sigma^2 \in \Big(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big).$$

For two independent normal samples, $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$; with a common variance, the pooled statistic $\frac{\bar{X}_1-\bar{X}_2-(\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}} \sim t(n_1+n_2-2)$, where $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$.

Proof. $\sqrt{n}(\bar{X}-\mu)/\sigma \sim N(0,1)$ is independent of $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ (Theorem 4.2); their ratio is $t(n-1)$. The intervals invert the probability statements, e.g. $P(\chi^2_{1-\alpha/2} \le (n-1)S^2/\sigma^2 \le \chi^2_{\alpha/2}) = 1-\alpha$. For two samples, the chi-squares are independent and add. ■

More generally, in the one-way model with independent $X_{ij} \sim N(\mu_i, \sigma^2)$, $i = 1,\dots,k$, $j = 1,\dots,n_i$, the group means are independent of the within-group sum of squares $\sum_{i,j}(X_{ij}-\bar{X}_i)^2/\sigma^2 \sim \chi^2(n-k)$. Also $\sum_i n_i(\bar{X}_i - \bar{X} - (\mu_i-\bar{\mu}))^2/\sigma^2 \sim \chi^2(k-1)$ (proved via Theorem 4.9 below).

4.3 Order statistics

Let $X_1,\dots,X_n$ be a sample from a continuous pdf f with cdf F, and $X_{(1)} < \cdots < X_{(n)}$ the order statistics (ties have probability 0).

Theorem 4.5

The joint pdf of $(X_{(1)},\dots,X_{(n)})$ is $n!\, f(y_1)\cdots f(y_n)$ on $y_1 < \cdots < y_n$.

Proof. The map from $\mathbf{x}$ (with distinct coordinates) to its sorted version is $n!$-to-one. On each of the $n!$ regions $\{x_{\pi 1} < \cdots < x_{\pi n}\}$ it is a permutation of coordinates with $|J| = 1$. Apply Theorem 4.1 and add the $n!$ equal terms. ■

Theorem 4.6

$$f_{X_{(r)}}(y) = \frac{n!}{(r-1)!(n-r)!}F(y)^{r-1}f(y)\{1-F(y)\}^{n-r},$$

and for $r < s$,

$$f_{X_{(r)},X_{(s)}}(y,z) = \frac{n!}{(r-1)!(s-r-1)!(n-s)!} \times F(y)^{r-1}f(y)\{F(z)-F(y)\}^{s-r-1}f(z)\{1-F(z)\}^{n-s}, \quad y < z.$$

Proof. $X_{(r)} \in (y, y+dy]$ requires $r-1$ observations below y, one in $(y, y+dy]$ and $n-r$ above: a multinomial event with probability $\frac{n!}{(r-1)!1!(n-r)!}F^{r-1}(f\,dy)(1-F)^{n-r}$. The joint case uses five cells in the same way. (Alternatively integrate Theorem 4.5.) ■

In particular $X_{(n)}$ has cdf F^n and $X_{(1)}$ has cdf $1-(1-F)^n$. Also $P(X_{(r)} \le y) = P(\#\{X_i \le y\} \ge r) = \sum_{k \ge r}\binom{n}{k}F(y)^k(1-F(y))^{n-k}$.

Example 4.4

For a $U(0,1)$ sample, $X_{(r)} \sim \text{Beta}(r, n-r+1)$, so $EX_{(r)} = \frac{r}{n+1}$. Since $F(X_i) \sim U(0,1)$ for any continuous F, $F(X_{(r)}) \sim \text{Beta}(r, n-r+1)$ always, which underlies distribution-free intervals for quantiles. For $\text{Exp}(\beta)$ samples, $X_{(1)} \sim \text{Exp}(\beta/n)$. More generally the spacings $(n-i+1)(X_{(i)} - X_{(i-1)})$ are i.i.d. $\text{Exp}(\beta)$; this follows from Theorem 4.5 with a linear change of variables of unit Jacobian after scaling.

The **sample median** is $X_{((n+1)/2)}$ (n odd) and the **range** is $X_{(n)} - X_{(1)}$; their distributions follow from Theorem 4.6 by a further change of variables.

4.4 The multivariate normal distribution

Definition (Multivariate normal)

$\mathbf{X} \sim N_n(\mu, \Sigma)$ if $\mathbf{X} \overset{d}{=} A\mathbf{Z} + \mu$ for some $n \times m$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_m(\mathbf{0}, I)$ (i.i.d. standard normals). Equivalently, its mgf is $M(\mathbf{t}) = \exp(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$.

Theorem 4.7

(a) $E\mathbf{X} = \mu$, $\operatorname{Var}\mathbf{X} = \Sigma$, and the distribution depends only on (μ, Σ). (b) If Σ is nonsingular,

$$f(\mathbf{x}) = \det(2\pi\Sigma)^{-1/2} \exp\{-\tfrac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\}.$$

(c) $B\mathbf{X} + \mathbf{c} \sim N(B\mu + \mathbf{c}, B\Sigma B')$; in particular every marginal and every linear combination is normal. (d) Components (or subvectors) are independent iff they are uncorrelated.

Proof. (a) $M(\mathbf{t}) = Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu} M_{\mathbf{Z}}(A'\mathbf{t}) = \exp(\mathbf{t}'\mu + \frac{1}{2}\mathbf{t}'AA'\mathbf{t})$, which involves only μ and Σ; the cgf gives the mean and covariance. (b) With $A = \Sigma^{1/2}$ (square, invertible), apply Theorem 4.1 to $\mathbf{x} = A\mathbf{z} + \mu$: $|J| = \det(A)^{-1}$ and $\mathbf{z}'\mathbf{z} = (\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$. (c) $B\mathbf{X} + \mathbf{c} = (BA)\mathbf{Z} + (B\mu + \mathbf{c})$. (d) If $\Sigma = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$, the mgf factors into the mgfs of the two subvectors. ■

Independence-iff-uncorrelated is special to *jointly* normal vectors: if $Z \sim N(0,1)$ and $W = \pm Z$ with an independent fair sign, W is normal and uncorrelated with Z but not independent of it.

Theorem 4.8 (Conditional distributions)

Partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$ with Σ_{22} nonsingular. Then

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \;\sim\; N\big(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \mu_2),\; \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\big).$$

Proof. Let $\mathbf{W} = \mathbf{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$. Then $\operatorname{Cov}(\mathbf{W}, \mathbf{X}_2) = \Sigma_{12} - \Sigma_{12} = 0$, so $\mathbf{W}$ is independent of $\mathbf{X}_2$ by Theorem 4.7(d). Its law is $N(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$. Write $\mathbf{X}_1 = \mathbf{W} + \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$ and condition on $\mathbf{X}_2 = \mathbf{x}_2$. ■

For a bivariate normal: $E(Y \mid X = x) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)$ is linear, and $\operatorname{Var}(Y \mid X) = \sigma_2^2(1-\rho^2)$ is constant.

Theorem 4.9 (Quadratic forms)

(a) If $\mathbf{X} \sim N_k(\mu, \Sigma)$ with Σ nonsingular, then $(\mathbf{X}-\mu)'\Sigma^{-1}(\mathbf{X}-\mu) \sim \chi^2(k)$. (b) If $\mathbf{Z} \sim N_n(\mathbf{0}, I)$ and A is symmetric idempotent ($A^2 = A$) of rank r, then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$. (c) If moreover $AB = 0$ for symmetric idempotent B, then $\mathbf{Z}'A\mathbf{Z}$ and $\mathbf{Z}'B\mathbf{Z}$ are independent.

Proof. (a) $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \mu) \sim N_k(\mathbf{0}, I)$ and the form equals $\mathbf{Z}'\mathbf{Z}$. (b) Diagonalize $A = P' \operatorname{diag}(I_r, 0) P$ with P orthogonal (the eigenvalues of an idempotent matrix are 0 or 1); $\mathbf{Y} = P\mathbf{Z} \sim N(\mathbf{0}, I)$ and $\mathbf{Z}'A\mathbf{Z} = Y_1^2 + \cdots + Y_r^2$. (c) $A\mathbf{Z}$ and $B\mathbf{Z}$ are jointly normal with $\operatorname{Cov} = AB' = AB = 0$, hence independent; and $\mathbf{Z}'A\mathbf{Z} = \|A\mathbf{Z}\|^2$, $\mathbf{Z}'B\mathbf{Z} = \|B\mathbf{Z}\|^2$. ■

Theorem 4.2 is the special case $A = I - \frac{1}{n}\mathbf{1}\mathbf{1}'$ (rank $n-1$), $B = \frac{1}{n}\mathbf{1}\mathbf{1}'$. The one-way-model statement after Corollary 4.4 follows the same way, with the projection onto group means minus the projection onto the grand mean (rank $k-1$).

Chapter 5. Approximating Sampling Distributions

Aim

Exact sampling distributions (Chapter 4) are rarely available. This chapter develops the large-sample tools that replace them: convergence in distribution and in probability, the central limit theorem, Slutsky's theorem and the delta method. It ends with simulation.

5.1 The central limit theorem

Theorem 5.1 (Central limit theorem)

Let $X_1, X_2, \dots$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Then for every x,

$$P\Big(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \le x\Big) \to \Phi(x).$$

Proof assuming an mgf. Let $Y_i = (X_i - \mu)/\sigma$ with mgf M_Y near 0; then $M_Y(t) = 1 + \frac{t^2}{2} + o(t^2)$. The mgf of $Z_n = \frac{1}{\sqrt{n}} \sum Y_i$ is $M_Y(t/\sqrt{n})^n = \big(1 + \frac{t^2}{2n} + o(\frac{1}{n})\big)^n \to e^{t^2/2}$, the $N(0,1)$ mgf. Conclude by the continuity theorem (Theorem 5.2). Without an mgf the same argument works with characteristic functions. ■

- **Normal approximation to the binomial** (de Moivre–Laplace): $X \sim B(n,p)$ is a sum of Bernoullis, so $(X - np)/\sqrt{np(1-p)} \xrightarrow{d} N(0,1)$. For integer x use the continuity correction $P(X \le x) \approx \Phi\big((x + \frac{1}{2} - np)/\sqrt{np(1-p)}\big)$. Similarly for Poisson($\lambda$) as $\lambda \to \infty$, and for $\chi^2(r)$ as $r \to \infty$: $(\chi^2(r) - r)/\sqrt{2r} \xrightarrow{d} N(0,1)$.
- **Multivariate CLT.** For i.i.d. random vectors with mean μ and covariance Σ, $\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \xrightarrow{d} N_k(\mathbf{0}, \Sigma)$. For multinomial counts, $(\mathbf{X}_n - n\mathbf{p})/\sqrt{n} \xrightarrow{d} N_k(\mathbf{0}, \mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}')$, a singular normal since the counts sum to n.

Example 5.1

For $X \sim B(100, 0.5)$: $P(X \le 55) \approx \Phi\big(\frac{55.5-50}{5}\big) = \Phi(1.1) = 0.864$ (exact: 0.864); without the correction, $\Phi(1) = 0.841$.

5.2 Limiting distributions and convergence in probability

Definition (Convergence in distribution)

$X_n \xrightarrow{d} Z$ if $P(X_n \le x) \to P(Z \le x)$ at every continuity point x of the cdf of Z. The law of Z is the **limiting** (asymptotic) distribution.

Continuity points matter. If X_n is uniform on $(0, 1 + \frac{1}{n})$ with probability $\frac{1}{2}$ and equals $1 + \frac{1}{n}$ otherwise, the cdfs converge to $\frac{x}{2}$ on $[0,1)$ and to 1 for $x > 1$, but at $x = 1$ the limit is $\frac{1}{2}$. The natural limit Z (uniform on $(0,1)$ with probability $\frac{1}{2}$, the value 1 otherwise) has cdf 1 at 1, and $X_n \xrightarrow{d} Z$ even though convergence fails at that one point.

Theorem 5.2 (Continuity theorem)

If $M_{X_n}(t) \to M_Z(t)$ for all t in a neighborhood of 0, then $X_n \xrightarrow{d} Z$. (Proof omitted; it is the mgf analogue of Lévy's theorem.)

Definition (Convergence in probability)

$X_n \xrightarrow{p} c$ (written $\operatorname{plim} X_n = c$) if $P(|X_n - c| \geq \epsilon) \to 0$ for every $\epsilon > 0$. For vectors, use the norm $\|\mathbf{X}_n - \mathbf{c}\|$.

Theorem 5.3

(a) $X_n \xrightarrow{p} c$ iff $X_n \xrightarrow{d} c$ (the constant). (b) If $E(X_n - c)^2 \to 0$ then $X_n \xrightarrow{p} c$. (c) A vector converges in probability iff each coordinate does.

Proof. (a) The cdf of the constant c is continuous except at c. If $X_n \xrightarrow{p} c$, then for $x < c$, $P(X_n \leq x) \leq P(|X_n - c| \geq c - x) \to 0$, and for $x > c$, $P(X_n \leq x) \geq 1 - P(|X_n - c| \geq x - c) \to 1$. Conversely $P(|X_n - c| \geq \epsilon) \leq P(X_n \leq c - \epsilon) + 1 - P(X_n \leq c + \epsilon/2) \to 0$. (b) Markov's inequality with $r = 2$. (c) $|X_{ni} - c_i| \leq \|\mathbf{X}_n - \mathbf{c}\| \leq \sum_i |X_{ni} - c_i|$. ■

Theorem 5.4 (Weak law of large numbers)

If X_i are i.i.d. with $E|X_1| < \infty$, then $\bar{X}_n \xrightarrow{p} EX_1$.

Proof under finite variance. $E(\bar{X}_n - \mu)^2 = \sigma^2/n \to 0$; apply Theorem 5.3(b). (The finite-mean case uses truncation or characteristic functions.) ■

So sample moments are consistent: $\frac{1}{n}\sum X_i^r \xrightarrow{p} EX_1^r$ whenever $E|X_1|^r < \infty$.

Theorem 5.5 (Continuous mapping)

If $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$ and g is continuous at $\mathbf{c}$, then $g(\mathbf{X}_n) \xrightarrow{p} g(\mathbf{c})$. In particular plim commutes with $+$, $-$, $\times$, and with $\div$ when the denominator's limit is nonzero.

Proof. Given $\epsilon > 0$ pick δ with $\|\mathbf{x} - \mathbf{c}\| < \delta \Rightarrow |g(\mathbf{x}) - g(\mathbf{c})| < \epsilon$. Then $P(|g(\mathbf{X}_n) - g(\mathbf{c})| \geq \epsilon) \leq P(\|\mathbf{X}_n - \mathbf{c}\| \geq \delta) \to 0$. For the arithmetic rules, $(X_n, Y_n) \xrightarrow{p} (a, b)$ by Theorem 5.3(c), and $x + y$, xy, x/y are continuous there. ■

Hence $S_n^2 = \frac{n}{n-1}(\frac{1}{n}\sum X_i^2 - \bar{X}_n^2) \xrightarrow{p} \sigma^2$ and $S_n \xrightarrow{p} \sigma$.

5.3 Computing limiting distributions

Theorem 5.6 (Slutsky)

If $X_n \xrightarrow{d} Z$ and $Y_n \xrightarrow{p} c$, then (a) $X_n + Y_n \xrightarrow{d} Z + c$; (b) $Y_n X_n \xrightarrow{d} cZ$; (c) $X_n / Y_n \xrightarrow{d} Z/c$ if $c \neq 0$.

Proof. (a) For a continuity point z of $Z + c$ and small ϵ such that $z \pm \epsilon$ are continuity points,

$$P(X_n + Y_n \leq z) \leq P(X_n \leq z - c + \epsilon) + P(|Y_n - c| \geq \epsilon),$$

and similarly $P(X_n + Y_n \leq z) \geq P(X_n \leq z - c - \epsilon) - P(|Y_n - c| \geq \epsilon)$. Let $n \to \infty$, then $\epsilon \downarrow 0$. (b) First $c = 0$: $P(|Y_n X_n| \geq \epsilon) \leq P(|X_n| > k) + P(|Y_n| \geq \epsilon/k)$; the first term is small for large

k uniformly in large n (tightness of X_n), the second $\to 0$. In general $Y_nX_n = (Y_n - c)X_n + cX_n$; apply the case $c = 0$ and (a). (c) $1/Y_n \xrightarrow{p} 1/c$ by Theorem 5.5; apply (b). ■

Example 5.2 (Studentized mean)

For any population with $0 < \sigma^2 < \infty$, $\frac{\sqrt{n}(\bar{X}_n-\mu)}{S_n} = \frac{\sqrt{n}(\bar{X}_n-\mu)/\sigma}{S_n/\sigma} \xrightarrow{d} N(0,1)$. Hence $\bar{X}_n \pm z_{\alpha/2}S_n/\sqrt{n}$ is an asymptotic $1-\alpha$ interval for μ, without normality. Under normality $t(n-1) \to N(0,1)$ is a special case.

Theorem 5.7 (Delta method)

If $\sqrt{n}(X_n-\theta) \xrightarrow{d} N(0,\sigma^2)$ and g is differentiable at θ, then $\sqrt{n}(g(X_n)-g(\theta)) \xrightarrow{d} N(0, g'(\theta)^2\sigma^2)$. In k dimensions, if $\sqrt{n}(\mathbf{X}_n - \theta) \xrightarrow{d} N(\mathbf{0}, \Sigma)$, then $\sqrt{n}(g(\mathbf{X}_n) - g(\theta)) \xrightarrow{d} N(0, \nabla g(\theta)'\Sigma\nabla g(\theta))$.

Proof. By differentiability, $g(x) - g(\theta) = (g'(\theta) + r(x))(x - \theta)$ with $r(x) \to 0$ as $x \to \theta$ ($r(\theta) := 0$). Since $X_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(X_n - \theta) \xrightarrow{p} 0$ by Slutsky, $r(X_n) \xrightarrow{p} 0$. Then $\sqrt{n}(g(X_n) - g(\theta)) = (g'(\theta) + r(X_n))\sqrt{n}(X_n - \theta) \xrightarrow{d} g'(\theta)N(0,\sigma^2)$ by Slutsky. The vector case is identical with a gradient. ■

Example 5.3

(a) For Bernoulli data, $\sqrt{n}(\hat{p}-p) \xrightarrow{d} N(0, p(1-p))$, so with $g(p) = \arcsin\sqrt{p}$, $g'(p) = \frac{1}{2\sqrt{p(1-p)}}$ gives $\sqrt{n}(\arcsin\sqrt{\hat{p}} - \arcsin\sqrt{p}) \xrightarrow{d} N(0, \frac{1}{4})$: a variance-stabilizing transformation. (b) For Poisson data, $2\sqrt{\bar{X}}$ has asymptotic variance $1/n$ whatever λ. (c) For the log-odds $\log\frac{\hat{p}}{1-\hat{p}}$, the asymptotic variance is $\frac{1}{np(1-p)}$.

Theorem 5.8 (Sample quantiles)

Let F have density f continuous and positive at $\xi_\alpha = F^{-1}(\alpha)$, and let $r_n/n \to \alpha$ with $\sqrt{n}(r_n/n - \alpha) \to 0$. Then $\sqrt{n}(X_{(r_n)} - \xi_\alpha) \xrightarrow{d} N(0, \frac{\alpha(1-\alpha)}{f(\xi_\alpha)^2})$.

Proof. $P(X_{(r_n)} \le \xi_\alpha + t/\sqrt{n}) = P(\#\{X_i \le \xi_\alpha + t/\sqrt{n}\} \ge r_n)$. The count is $B(n, p_n)$ with $p_n = F(\xi_\alpha + t/\sqrt{n}) = \alpha + f(\xi_\alpha)t/\sqrt{n} + o(n^{-1/2})$. By the CLT for binomials with $p_n \to \alpha$, the probability tends to $P(Z \ge \frac{-f(\xi_\alpha)t}{\sqrt{\alpha(1-\alpha)}}) = \Phi(\frac{f(\xi_\alpha)t}{\sqrt{\alpha(1-\alpha)}})$. ■

For the median of $N(\mu,\sigma^2)$, $\alpha = \frac{1}{2}$ and $f(\mu) = 1/(\sigma\sqrt{2\pi})$, so the asymptotic variance is $\pi\sigma^2/(2n)$, larger than σ^2/n for $\bar{X}$.

5.4 Approximation by simulation

When neither exact nor asymptotic theory is convenient, the distribution of a statistic T can be approximated by **Monte Carlo**: generate B independent samples from the model, compute $T_1, \ldots, T_B$, and use their empirical distribution.

- **Generating random numbers.** From $U \sim U(0,1)$, $F^{-1}(U) \sim F$ (Section 4.1). For example $-\beta\log(1-U) \sim \text{Exp}(\beta)$, $\log\frac{U}{1-U}$ is standard logistic, $\mu + \sigma\Phi^{-1}(U) \sim N(\mu,\sigma^2)$. Sums, ratios and so on of generated variables give gamma, χ^2, t, F variates.
- **Accuracy.** An estimated probability $\hat{p} = \frac{1}{B}\sum I(T_b \in A)$ is binomial, with standard error $\sqrt{p(1-p)/B}$. An estimated mean $\bar{T}$ has standard error $s_T/\sqrt{B}$. By the CLT, $\hat{p} \pm$

$2\sqrt{\hat{p}(1-\hat{p})/B}$ is an approximate 95% band for the simulation error; $B = 10^4$ gives about ± 0.004 for $p \approx 0.05$.

- **Uses.** Checking the accuracy of asymptotic approximations at a given n (e.g. coverage of $\bar{X} \pm z_{\alpha/2} S/\sqrt{n}$ for skewed populations), and approximating quantiles of statistics with intractable distributions.